https://www.youtube.com/watch?v=e-tpHjCg3xM

TSMC's Newcomer Training Center

TSMC revealed its Newcomer Training Center for the first time, located at the Fab 15A site in the Central Taiwan Science Park. New employees train there for about two months before starting real work.

The training covers four main areas: equipment, process, systems, and shift-duty skills. Employees also attend a separate training program covering things like water treatment and electrical systems, since the factory runs 24 hours a day.

One special feature is the use of virtual reality (VR). Employees use VR headsets to practice handling emergencies, such as fires or gas leaks, so they can learn how to react safely without needing a real emergency to happen first.

By late 2022, more than 6,300 new employees had completed the training, including staff from TSMC's plants in the US and Japan. The goal of the center is to help new employees adapt to real work faster and close the gap between what they learn in school and what the job actually requires.

My project is similar to TSMC's training center in a few ways.

Both focus on hands-on training, not just classroom learning. TSMC helps new employees learn real job skills quickly; my project does this too, by working with schools so students can join workshops and internships.

Both are located near the Science Park, making it easy for workers to get to their jobs.

The courses are similar — TSMC teaches equipment, process, and systems skills, while my project covers semiconductor technology and workplace skills.

-https://www.bnext.com.tw/article/72635/tsmc-nyc-facilityacademy

According to the BusinessNext report, the training steps are as follows:

1. Straight to the training center after being hired

Production-line staff (engineers and assistant engineers) don't go straight to the fab

after being hired. Instead, they first spend 4 to 8 weeks training at TSMC's Newcomer Training Center in the Central Taiwan Science Park.

2. Training divided by process area
Inside the center, training is organized by process area — such as thin film, wet etching, and dry etching — with new hires trained in the classroom matching their assigned area.

3. Hands-on practice with real equipment
The report specifically notes that every classroom contains the actual machines used on the production line. In the key photolithography (yellow light) area, for example, there's a real machine for trainees to operate, and even a wafer transport cart like the ones used inside the fab.

4. Learning from disassembled modules
Classrooms also contain disassembled equipment modules, letting trainees see the internal structure and understand how the machines actually work.

5. Internal cameras for full visibility
Since most of the internal workings of production machines can't normally be seen, TSMC installed cameras inside the equipment so trainees can clearly watch what happens to a wafer once it enters the machine. One instructor is quoted as saying: "Here, what we teach is the process from zero to one."

6. Smoother transition to the real job
After finishing training at the center, new hires return to the fab already equipped with foundational knowledge, making it easier to communicate with senior colleagues and adapt to the actual work.

- TSMC Careers — Equipment Engineer (Etch): https://careers.tsmc.com/en_US/careers/JobDetail/Equipment-Engineer-Etch/16963

- TSMC Careers — Intelligent Manufacturing Engineer: https://careers.tsmc.com/en_US/careers/JobDetail/Intelligent-Manufacturing-Engineer-IMC-MFG/5439
- TSMC Careers — 2026 Campus Recruitment (Intelligent Manufacturing): https://careers.tsmc.com/en_US/careers/JobDetail/2026-TSMC-Campus-Recruitment-Intelligent-Manufacturing-Engineer-IMC-MFG/19040
- [redacted] https://www.ziprecruiter.com/Jobs/Tsmc-Engineer